Muhammad Umair Danish

Postdoctoral Associate / ML Research Engineer

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Introduction

My research designs and develops machine-learning and interpretability methods, such as mechanistic interpretability and human-centered explainable AI, and specialized machine learning, such as physics-guided deep learning and time-series forecasting, for energy, cyber-physical systems, environmental, financial, and medical applications. I have 22 peer-reviewed publications (11 journal, 11 conference), including IEEE Transactions and CVPR. Publications are ordered by venue standing. In the lists below, my name is in bold and '*' marks the corresponding author. Impact factors (IF) are from Journal Citation Reports and CiteScores from Scopus; citation counts are from Google Scholar as of October 2026.

Academic Experience

1/2026 - Present **Western University, London, Canada**
Postdoctoral Associate
Recipient of the prestigious Western Postdoctoral Fellowship.
Advised by Katarina Grolinger.

1/2023 - 12/2025 **Western University, London, Canada**
Ph.D., Electrical and Computer Engineering
Advised by Katarina Grolinger.

6/2022 - 12/2022 **Gwangju Institute of Science and Technology, Gwangju, South Korea**
Machine Learning Researcher - Medical Image Registration
Bio-Medical Information & Signal (BMIS) Lab.

2/2018 - 12/2019 **The University of Lahore, Lahore, Pakistan**
M.S., Software Engineering
Gold Medalist.
Advised by Muhammad Sultan Zia.

1/2013 - 7/2018 **COMSATS University Islamabad, Pakistan**
B.S., Software Engineering
Roll of Honor.
Advised by Muzammil Behzad.

Honors & Awards

2025 **Western Postdoctoral Fellowship, Western University**
Prestigious university-wide award funded by the Vice-President (Research). Listed among the 2025 recipients.

2025 **Top 1% Cited Paper, Scopus**
Scopus 99th citation percentile (FWCI 6.22) for "Kolmogorov–Arnold Recurrent Network for Short-Term Load Forecasting Across Diverse Consumers".

2025 **Best Paper Award, IEEE AIxSET**
Awarded to the paper "Distributional Feature Separability for Explainability of Neural Networks for Financial Credit Evaluation".

2024 **Western Graduate Fellowship, Western University**

2023 **Western Graduate Research Scholarship, Western University**

2022 **Korean Government Scholarship**

2019 **Gold Medal, The University of Lahore**
Awarded for first position in M.S. Software Engineering.

2018 **Roll of Honor, COMSATS University Islamabad**
Awarded in B.S. Software Engineering.

Journal Publications

1. **Danish, M. U.**; Grolinger, K. Leveraging Hypernetworks and Learnable Kernels for Consumer Energy Forecasting Across Diverse Consumer Types. *IEEE Transactions on Power Delivery*, 40(1), 75–87, Feb. 2025. [IF 4.7 · CiteScore 11.2 · 24 citations]

2. **Danish, M. U.**; Ali, K.; Siddiqui, K.; Grolinger, K. Physics-Guided Memory Network for Building Energy Modeling. *Energy and AI*, 21, 100538, Sept. 2025. [IF 9.7 · CiteScore 16.3 · 12 citations]

3. Buwaneswaran, M.; **Danish, M. U.**; Grolinger, K. Conditional Denoising Diffusion Model for Missing Value Imputation in Smart Meter Data. *Engineering Applications of Artificial Intelligence*, 181, 115403, Oct. 2026. [IF 9.0 · CiteScore 11.7 · 1 citation]

4. Aziz, M.; Rehman, U.; **Danish, M. U.**; Grolinger, K. Global-Local Image Perceptual Score (GLIPS): Evaluating Photorealistic Quality of AI-Generated Images. *IEEE Transactions on Human-Machine Systems*, 55(2), 223–233, Apr. 2025. [IF 4.4 · CiteScore 10.6 · 40 citations]

5. **Danish, M. U.**; Grolinger, K. Kolmogorov–Arnold Recurrent Network for Short-Term Load Forecasting Across Diverse Consumers. *Energy Reports*, 13, 713–727, 2025. [IF 6.6 · CiteScore 15.3 · 65 citations · **Top 1% cited — Scopus 99th percentile · FWCI 6.22**]

6. **Danish, M. U.**; Aziz, M.; Rehman, U.; Grolinger, K. Linear Lens: A Human-Centered, Non-Interventional Mechanistic Approach to Explainable AI. *Machine Learning with Applications*, 24, 100897, June 2026. [IF 6.1 · CiteScore 12.6 · 0 citations]

7. Gholamiangonabadi, D.; **Danish, M. U.***; Chen, Z.; Grolinger, K. Autoencoder-Based Clustered Federated Learning with Importance-Aware Aggregation for Sentiment Analysis on Non-IID Data. *Machine Learning with Applications*, 26, 101046, 2026. [IF 6.1 · CiteScore 12.6 · 2 citations]

8. Aziz, M.; Rehman, U.; **Danish, M. U.**; Ali, S.; Abbasi, A. Z. Towards a Unified Evaluation Framework: Integrating Human Perception and Metrics for AI-Generated Images. *Multimedia Systems*, 31(3), 180, Apr. 2025. [IF 2.8 · CiteScore 6.7 · 22 citations]

9. **Danish, M. U.**; Rehman, U.; Grolinger, K. Monotone Delta: An Order-Theoretic Tournament Graph Approach for Internal Consistency Assessment. *Frontiers in Applied Mathematics and Statistics*, 12, 1687194, Mar. 2026. [IF 2.0 · CiteScore 3.6 · 4 citations]

10. Kogutek, D.; Carroll, L.; **Danish, M. U.**; Lu, C. Dispersity Measures Within Sessions During Improvised Active Music Therapy with Clients with Parkinson's Disease. *Approaches: An Interdisciplinary Journal of Music Therapy*, 2026. *Accepted for publication.*

11. Kogutek, D.; **Danish, M. U.** MidiPy: A Python Application for Processing MIDI Data in Improvised Active Music Therapy. *Journal of Creative Music Systems*, 2026. *Accepted.*

Conference Publications

1. Tissera, D.; Awadallah, O.; **Danish, M. U.**; Sadhu, A.; Grolinger, K. Any-Class Presence Likelihood for Robust Multi-Label Classification with Abundant Negative Data. In IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Denver, CO, USA, June 2026, pp. 2429–2439. [2 citations]

2. **Danish, M. U.**; Aziz, M.; Rehman, U.; Grolinger, K. Cognitive Effects of Textual, Graphical, and Interactive Explanations in Explainable Artificial Intelligence. In 52nd Annual Conference of the IEEE Industrial Electronics Society (IECON 2026), Doha, Qatar, Oct. 2026. *Accepted.*

3. **Danish, M. U.**; Aziz, M.; Rehman, U.; Zecevic, A.; Grolinger, K. Human-Centered Explainable Artificial Intelligence and the Effects of Explanation Modalities on Trust, Understanding, and Decision Making. In IECON 2026, Doha, Qatar, Oct. 2026. *Accepted.*

4. **Danish, M. U.**; Buwaneswaran, M.; Fonseka, T.; Grolinger, K. Graph Attention Convolutional U-Net: A Semantic Segmentation Model for Identifying Flooded Areas. In 50th Annual Conference of the IEEE Industrial Electronics Society (IECON 2024), Chicago, IL, USA, Nov. 2024. [18 citations]

5. **Danish, M. U.**; Noorchenarboo, M.; Narayan, A.; Grolinger, K. ChebyRegNet: An Unsupervised Deep Learning Technique for Deformable Medical Image Registration. In IECON 2024, Chicago, IL, USA, Nov. 2024, pp. 1–8. [3 citations]

6. **Danish, M. U.**; Aziz, M.; Rehman, U.; Grolinger, K. How Explanation Intent Shapes Human Use of XAI for Neural-Network Predictions. In 2026 IEEE 8th International Conference on Cognitive Machine Intelligence (CogMI), San Jose, CA, USA, Nov. 2026. *Accepted, Application Track.*

7. **Danish, M. U.**; Yousaf, A. A.; Aziz, M.; Grolinger, K.; Rehman, U. RootSolver: A Cognitive-Inspired Approach to Polynomial Root-Finding with Multiplicity Detection. In 17th International Conference on Advanced Computational Intelligence (ICACI), Bath, UK, July 2025, pp. 243–250. [0 citations]

8. Aziz, M.; **Danish, M. U.**; Rehman, U. Distributional Feature Separability for Explainability of Neural Networks for Financial Credit Evaluation. In 2025 Conference on AI, Science, Engineering, and Technology (AIxSET), Laguna Hills, CA, USA, Oct. 2025, pp. 127–133. *Best Paper Award.* [0 citations]

9. Yousaf, A. A.; Aziz, M.; **Danish, M. U.**; Safi, S. A.; Rehman, U. Statistically-Guided Kernel Regression: Embedding Feature Relevance into Kernel Learning. In 2nd International Conference on Advanced Machine Learning and Data Science (AMLDS 2026), Osaka, Japan, July 2026. *Accepted*; proceedings pending. [0 citations]

10. Aziz, M.; **Danish, M. U.**; Grolinger, K.; Rehman, U.; Iqbal, F. Finite-Difference Approximation for Explaining Neural Network Predictions in Financial Credit Evaluation. In International Conference on Advanced Machine Learning and Data Science (AMLDS 2025), Tokyo, Japan, July 2025, pp. 12–18. [1 citation]

11. **Danish, M. U.**; Iqbal, F.; Fung, B. C. M.; Marrington, A. Interpretable by Design: Traceable Edge-Function Kolmogorov–Arnold Networks for Memory-Forensic Malware Detection. In 2026 6th International Conference on Artificial Intelligence and Application Technologies (AIAT 2026), Tokyo, Japan, Dec. 2026. *Accepted for publication and oral presentation.*

Monograph Publications

1. **Danish, M. U.** *Neural Networks and Their Interpretability for Building Energy Modeling*. Ph.D. dissertation, Western University, 2025.

2. **Danish, M. U.** *Image Denoising and Restoration Using Convolutional Neural Networks*. M.S. thesis, The University of Lahore, 2019.

3. **Danish, M. U.** *A Comparative Analysis of Image Restoration Techniques*. B.S. thesis, COMSATS University Islamabad, 2018.

Talks

1. *Deep learning architectures and mechanistic interpretability for energy forecasting*. Invited Guest Lecture, ECE9603, Western University. December 2025.

2. *Mechanistic interpretability for financial applications*. Research Talk, University of Toronto. October 2025.

3. *Non-interventional interpretability and explainability of neural networks*. Research Presentation, HFES Waterloo Chapter, University of Waterloo. June 2025.

4. *Time series forecasting and physics-informed machine learning*. CAAF-ECCC Research Day, Western University. June 2025.

5. *HyperForecasting: Leveraging kernelized hypernetworks for short term load forecasting across diverse consumers*. Graduate Research Symposium, Western University. June 2024.

6. *Challenges in medical image analysis using deep learning*. Research Seminar, Gwangju Institute of Science and Technology (GIST), South Korea. October 2022.

Research Experience

6/2023 - Present **Wilfrid Laurier University, Waterloo, Canada**
Research Assistant, Music Data Science and Software
Co-developed MidiPy, open-source software for MIDI feature extraction and visual analytics in music-therapy research.

5/2026 - 8/2026 **Western University, London, Canada**
Research Assistant (Postgrad), Human-Centered AI and HCI
Designed human–AI studies on explanation modality, trust, cognitive load, and appropriate reliance.

1/2026 - 10/2026 **Zayed University, Remote, United Arab Emirates**
Research Assistant, AI and Digital Forensics
Built reproducible pipelines over approximately 2.6 million malware records across Windows, Linux, and memory artifacts.

1/2023 - 12/2025 **Western University, London, Canada**
Graduate Research Assistant, Machine Learning
Developed hypernetwork, Kolmogorov–Arnold recurrent, and physics-guided memory network models for time-series forecasting.

Teaching

Fall 2025 **SE 3309: Database Management Systems, Western University**
Undergraduate software engineering course. Teaching assistant, 140 hours.

Winter 2025 **SE 3350: Software Engineering Design I, Western University**
Undergraduate software engineering course. Teaching assistant, 140 hours.

Fall 2024 **SE 3309: Database Management Systems, Western University**
Undergraduate software engineering course. Lead teaching assistant, 140 hours.

Summer 2024 **SE 2251: Software Design, Western University**
Undergraduate software engineering course. Teaching assistant, 13 hours.

Winter 2024 **SE 2251: Software Design, Western University**
Undergraduate software engineering course. Teaching assistant, 70 hours.

Fall 2023 **SE 3309: Database Management Systems, Western University**
Undergraduate software engineering course. Teaching assistant, 140 hours.

Winter 2023 **ENGSCI 1036: Programming Fundamentals for Engineers, Western University**
Undergraduate engineering course. Teaching assistant, 70 hours.

2021 **Pharmacy Practice-I: Pharmaceutical Mathematics and Biostatistics, The University of Lahore**
Pharm-D professional course. Main instructor.

2018 **Programming Fundamentals (C++), The University of Lahore**
Undergraduate B.S. Physics course, taught twice. Main instructor.

External Services

Present **Journal Reviewer**
IEEE Transactions on Industrial Informatics; *Engineering Applications of Artificial Intelligence*; *Scientific Reports*; *Electrical Engineering* (Springer); *Signal, Image and Video Processing* (Springer).

Leadership Experience

3/2021 - 5/2022 **GIFT University, Gujranwala, Pakistan**
Assistant Director, Quality Enhancement Cell
Led self-assessment, program review, and accreditation readiness for academic programs under Higher Education Commission quality standards.

3/2014 - 3/2021 **University of Chenab, Gujrat, Pakistan**
11/2021 - 5/2022 *Manager / Assistant Director, Quality Enhancement*
Formerly The University of Lahore, Gujrat Campus. Coordinated program self-assessments, stakeholder surveys, and quality audits.

Open-Source Software

1. Linear Lens: non-interventional neuron- and layer-level interpretability for PyTorch models, with in-browser verification on 45 public datasets and 180 trained-model analyses. umairdanish.com/software/linear-lens.
2. Core-Norm: robust, bounded, invertible normalization for numeric ML, with scikit-learn integration. github.com/udanish50.
3. MidiPy: MIDI parsing, feature extraction, and visual analytics. midipy-online.streamlit.app.
4. OpenMetricLab: local-first evaluation for regression, classification, segmentation, and image registration. github.com/udanish50.